

AirSense — Finalized Problem Statement

Urban Pollution Incident-Response System, reframed for India

Prepared for: Devesh Gujarathi | Basis: AirSense Competitive & Landscape Analysis | August 2026

Where We Started

“This neighbourhood is experiencing an abnormal pollution event. Here is the most likely cause, who or what should be investigated, what intervention is appropriate, and whether pollution actually decreased after the intervention.”

This framing assumed an ambient event and said nothing about who reports it, who acts on it, or what system it plugs into — all of which the competitive analysis shows is where the real, unaddressed gap sits.

Final Problem Statement (Submission-Ready)

Indian cities already have a channel for citizens to report pollution — CPCB's SAMEER app and equivalent state grievance portals generate thousands of complaints a month. But everything downstream of “complaint filed” is manual: no automated diagnosis of the likely source, no estimate of how far the incident has spread, no recommended action for the nodal officer who receives it, and no way to confirm whether the fix actually worked. **AirSense is an AI intelligence layer that plugs into this existing complaint infrastructure to close that loop** — for a reported incident, it infers the probable source (starting with construction/road dust), estimates a wind-adjusted affected zone, recommends an intervention to the responsible agency, and measures before/after AQI change to prove impact. Piloted in Pune using data already available from Oizom's Smart City sensors, CPCB's CAAQMS stations, and SAFAR's forecasts — with an architecture built to plug into any of the 130+ NCAP cities already running the same kind of grievance system, without needing new hardware.

One-Line / Pitch Version

“AirSense turns India's existing pollution complaints into accountable action — diagnosing the source, mapping the affected zone, recommending a fix, and proving whether it worked — for any city already running a SAMEER-style grievance system.”

Why This Is a Sharper Cut, Not Just a Rewrite

- **Triggered by real, existing behavior, not a hypothetical.** SAMEER already gets 3,000+ complaints in 10 days during active campaigns — the intake habit exists, the intelligence layer doesn't. You're completing something citizens already use, not asking them to adopt something new.
- **Closes the one gap literally nobody else closes.** Across 15 competitors in 6 categories, not one logs a measured before/after outcome per incident. That's the actual moat — the RF/XGBoost/SHAP stack is commodity technique anyone in this space could replicate.
- **Scalable by architecture, not by ambition.** Because it plugs into SAMEER/NCAP-style infrastructure that 130+ cities are already required to run, “expand to more cities” isn't a separate roadmap — it's the same integration pointed at a different city's grievance API.

- **Doesn't compete where it would lose.** Not trying to out-forecast SAFAR or out-cover Google's AQ API — both structurally stronger there, always will be. This builds the workflow layer every well-funded player has left conspicuously empty.
- **Falsifiable, which makes it defensible.** “Did AQI actually improve after this specific intervention” is a testable claim, demonstrable in a project defense or a government pitch — stronger footing than “our dashboard is accurate.”

Scope Path (Prove → Generalize → Scale)

1. Prove it

One incident type (construction/road dust), one city (Pune), using Oizom + CPCB + SAFAR data rather than new hardware.

2. Generalize it

Extend source-classification to vehicular, industrial, and biomass incidents once the pipeline works end-to-end.

3. Scale it

Offer the same integration to any NCAP city already running a SAMEER-style portal — the product doesn't change, only which city's grievance feed it's wired into.

This progression is what makes the statement both **smart** (evidenced, falsifiable, positioned against a documented gap) and **scalable** (the scaling mechanism is baked into which infrastructure it sits on top of, not something to invent later).

Source: AirSense Competitive & Landscape Analysis (Aug 2026) — 15 competitors mapped across 6 categories; problem statement reframed against the CPCB SAMEER / NCAP grievance-infrastructure whitespace identified in that analysis.